ARDOR / INDEPENDENT FUNCTION MODULE

Passive expression pedal / I2C ADC / HAND ASSEMBLY

T1 | HAND ASSEMBLY | SEE PIN GUIDE AT RIGHT

PIN GUIDE: IN = into this board / OUT = out / I/O = both / GND = 0 V

J302 / PASSIVE PEDAL
1: TIP I/O 2: RING I/O
3: GND 0V

JP301 / PEDAL JUMPER
1: TIP 2: RING
3: PEDAL IN 4: EXC OUT
5: RING 6: TIP

JP303 / DATA PULLUP
1: PULLUP 2: DATA I/O

J101 / PWR IN / I2C
1: 3V3 IN 2: GND 0V
3: DATA I/O 4: CLOCK IN

JP302 / ADDR JUMPER
1: LOW GND 2: SELECT
3: HIGH 3V3

JP304 / CLOCK PULLUP
1: PULLUP 2: CLOCK IN

U301.2: UNUSED ALERT – LEAVE OPEN

J302
PASSIVE PEDAL INPUT / EXCITATION OUTPUT
3 GND
2 EXP_RING
1 EXP_TIP

JP301
JUMPER ONLY: PEDAL POLARITY
EXP_TIP 1 2 EXP_RING
3 4
EXP_RING 5 6 EXP_TIP

R301
1kR / 1% 0.25W
+3V3_A — EXP_EXC

D303
BAT85S
+3V3_A

R302
10kR / 1% 0.25W

D304
BAT85S
GND

R303
1MR / 1% 0.25W
GND

C301
100n / 50V X7R
GND

R308
2.2kR / 1% 0.25W
+3V3_A

JP303
JUMPER ONLY: SDA PULLUP
1
SDA 2

R309
2.2kR / 1% 0.25W
+3V3_A

JP304
JUMPER ONLY: SCL PULLUP
1
SCL 2

D301
1.5KE6.8CA
CHASSIS
EXP_TIP

D302
1.5KE6.8CA
CHASSIS
EXP_RING

U301
ADS1115-M
+3V3_A
8 VDD
ALERT/RDY 2
4 AIN0
5 AIN1
6 AIN2
7 AIN3
10 SCL
9 SDA
GND ADDR 1
3 GND
GND
GND

R304
10kR / 1% 0.25W
EXP_EXC

D305
BAT85S
+3V3_A

C302
100n / 50V X7R
GND

R305
33R / 1% 0.25W
SCL

R306
33R / 1% 0.25W
SDA

J101
3V3 POWER INPUT + I2C TO HOST
+3V3 1
GND 2
3
4

JP302
JUMPER ONLY: ADC ADDRESS
GND 1
2
+3V3_A 3

D306
BAT85S
GND

R101
0R / 1% 0.25W
CHASSIS — GND

FB101
800R@100MHz / WE 7427501
+3V3 — +3V3_A

C101
10u / 25V polarized
GND

C102
100n / 50V X7R
GND

C303
100n / 50V X7R
+3V3_A
GND

R307
2.2kR / 1% 0.25W
+3V3_A
GND

Passive 10k–100k pot only. Sleeve = GND. JP301: both shunts 1–3 / 2–4 for tip wiper; 3–5 / 4–6 for ring wiper.
JP302: 1–2 = 0x48; 2–3 = 0x49. Never leave ADDR floating. JP303/304 enable local 2.2k I2C pull-ups.
Remove BOTH pull-up shunts when host already has pull-ups (Raspberry Pi). Start at 100 kHz I2C.
ADS1115: +/-4.096 V, 128 SPS. Read AIN0 and AIN1; normalize AIN0/AIN1, calibrate endpoints and smooth.

+3V3 +3V3_A CHASSIS GND

Engineering prototype – bench validation required

Ardor

Sheet: /
File: Ardor_Expression_THT.kicad_sch

Title: Passive expression pedal / I2C ADC / HAND ASSEMBLY

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